Vertical Mobility and RL-based Locomotion of a Snake-Like Robot for Building Foundation Inspection

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***Abstract*— Inspection of confined spaces in buildings, such as compartmentalized building foundations (pit spaces), is laborious and often hazardous for human workers. This paper presents Garden EELS, a new lightweight (7.5 kg), 1.83 m-long snake-like robot that can perform inspection tasks in place of human workers. The robot integrates (i) a leader–follower trajectory-based controller for vertical traversal of elevated access holes up to 92 cm in height, (ii) a reinforcement learning (RL)–based planar locomotion policy with structured sinusoidal parameterization and a command-integrated pose-tracking reward for robust velocity control, and (iii) onboard inspection capabilities including covariance-aware SLAM for 3D mapping and object detection. Field experiments conducted in representative underground pit spaces demonstrated reliable vertical traversal, sustained planar locomotion on concrete and gravel surfaces, and real-time mapping and defect detection. The results validate the feasibility of deploying compact snake robots for practical confined-space inspection, advancing robotic replacement of hazardous human labor in construction environments.**

I. INTRODUCTION

A. Background

Humans are still the main workforce in confined spaces in construction. For example, the foundations of large buildings in earthquake-prone areas, such as Japan, often consist of a grid of thick concrete walls, resulting in multiple compartmentalized underground spaces called *underground pit spaces*, as shown in Figure 1. Ensuring the integrity of the concrete foundations is crucially important. In addition, the pit spaces are often used for routing pipes or storing water. As such, they must be thoroughly inspected throughout the building lifecycle - during construction, before delivery, and during use. However, inspecting these spaces poses substantial risks to human workers. Many lives have been lost due to asphyxiation or exposure to toxic gases. Besides the hazards, it is an ergonomically poor environment due to dust, low ceilings, and obstacles such as running pipes. For these reasons, replacing human labor in underground pit spaces with robots is highly desired.

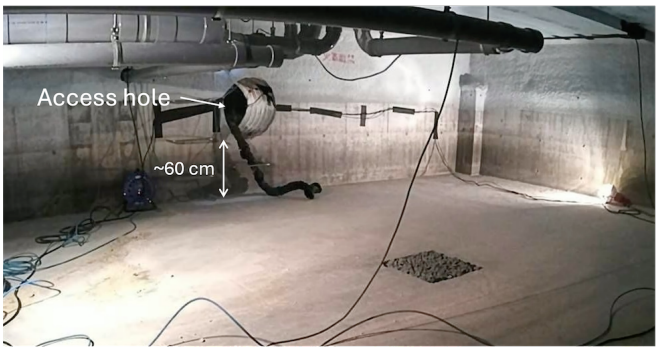


Fig. 1. The underground pit space used for the experiment and our lightweight snake-like Garden EELS robot traversing the elevated access hole. This pit space is representative of the state at building delivery. During construction, pit spaces are more cluttered with building materials, tools, and workers.

Robotic inspection of underground pit spaces is challenging due to dust, obstacles (pipes, debris, tools), poor lighting, poor accessibility, and the vastness and complexity of the space - with some of the largest buildings having hundreds to thousands of interconnected pit spaces. Adjacent pits are usually connected by a horizontal access hole, typically raised about 60 cm above the floor (Figure 1). This elevated geometry, designed to make it easier for human workers to pass through, is a non-trivial obstacle for robots. Furthermore, robots must be light enough ($\sim\leq$10 kg) to be carried and deployed by a single person for practicality and efficiency. This size limitation makes the 60 cm step even harder for robots to overcome. Wheeled vehicles without complex articulation are clearly unsuitable. Teleoperated drones were previously tested on an actual construction site by our sponsor, Kajima Corporation, but were found impractical due to their short flight duration and the need for constant radio communication.

We propose using snake-like robots as a solution. They have been proven viable for many applications in confined natural and artificial environments, such as crevasses [1], caves, and building pipes [2], [3]. They were utilized in actual disaster sites, such as the collapsed buildings after

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the 2017 Mexico City Earthquake [4] and the Fukushima Nuclear Power Plant [5]. The application of snake-like robots to confined-space inspection of buildings, in particular, the ability to clear large vertical steps ($\geq$ 30% of body length), is an underresearched topic with one single example [6]. Snake robots are viable for navigating through confined spaces with obstacles, and unlike drones, they can operate in dusty environments without kicking up clouds that obscure visual sensors. Previously, a family of snake-like robots, EELS [7], [8], demonstrated fully autonomous vertical mobility in natural holes on a glacier [1]. However, weighing 50–80 kg, these robots are not deployable by a single person and do not meet the aforementioned requirement. This paper presents a new lightweight snake-like robot, *Garden EELS*, a small EELS variant weighing 7.5 kg and measuring 1.8 m long. In this paper we review the design of the robot and demonstrate its capability through a successful demonstration in an underground pit located in Kajima Corporation's training facility. This demonstration involved 3D mobility through a 60 cm elevated access hole, horizontal mobility on the pit space floor using reinforcement learning, and wall inspection using a RealSense L515 Lidar/camera. We furthermore demonstrate the robot's climbing capability through laboratory experiments where it successfully climbed a 92-cm-high step (50% of its body length) using roll-correction controls.

B. Related Work

The majority of existing work on snake-like robots is concerned with mobility on near-horizontal surfaces, where obstacles are avoided sideways [9]. There are, of course, many successful examples of vertical mobility, such as climbing ladders [10] and poles [11]. While step climbing has been solved by snake-like robots with various configurations, the vast majority only deal with relatively low steps with heights equivalent to one or two module lengths, typically 20–30 cm [12], [13], [14], [15]. Only a handful of snake-like robots have achieved the feat of ascending to the top of a large step. The ACM-R8 robot by Komura et al. [16], which consists of five 30 cm-diameter mono-thread wheels, successfully climbed a 60 cm step, but it is too heavy (38 kg) for our purpose. The T^2 Snake-3 robot by Tanaka et al. [17], featuring a pair of active and passive wheels at every module and weighing only 9.2 kg, can climb up to a 1 m step. It uses a leader-follower gait involving an inchworming motion to avoid the edge, informed by proximity and distance sensors on every module. The same group used a similar robot for underground pit space inspection, including traversing a 60-cm high access hole [6]. Fu et al. [18] argued that compliance is key for maintaining stability during step climbing, and developed a 2.4 kg snake robot with a pair of passive one-way wheels with compliant suspension at every module. The robot climbed steps up to 41 cm in height. Interestingly, the three robots mentioned above are all equipped with wheels on every module, likely to mimic the anisotropic friction of biological snake skin, which is deemed critical for snake mobility [19]. In this paper, we explored an alternative, minimalist robot configuration that does not have contact sensors or wheels along the body. The majority of propulsion force comes from the friction between the body and the ground, assisted by an anisotropic bumper at every module and two pairs of passively transformable PaTS wheels [20]. With this configuration, the robot demonstrated the ability to climb steps up to 92 cm in height using simple open-loop controls.

Recent work has demonstrated the effectiveness of reinforcement learning for generating locomotion strategies in snake robots operating in complex and contact-rich environments. Zhang et al. [21] propose a reinforcement learning framework that enables snake robots to navigate a cluttered environment by learning motion policies directly from interaction with the environment. Additionally, sinusoidal gait bases have been widely shown to capture serpentine locomotion patterns while still permitting a rich class of turning and forward locomotion behaviors [22]. We introduce a structured action parameterization based on a spatiotemporal sinusoidal prior to reduce redundancy in gait representation. By restricting the policy to learn only amplitude modulations of predefined sinusoids, our method improves the identifiability of the control policy, reduces variance in gradient estimates, and decreases the dimensionality of the learning problem. This formulation complements existing reinforcement learning approaches by incorporating domain-specific structure directly into the policy representation, enabling efficient learning of stable locomotion strategies. Furthermore, we introduce a unicycle-integrated pose-tracking reward as an alternative to the widely used velocity-tracking reward, which is difficult to apply to a snake robot due to its undulating motions.

The novel contribution of this work is in the integrated system rather than the individual technical components. By adding anisotropic bumpers along the body and extremely grippy PaTS wheels on the tail and head, we enable climbing of high walls with fewer actuators and simpler control schemes that do not require proprioceptive feedback. This reduction in hardware requirements produces a lower-cost platform more suitable for large-scale commercial deployment.

II. GARDEN EELS ROBOT OVERVIEW

Garden EELS (Fig. 2), measuring 1.83 m long and weighing 7.5 kg, consists of 22 actuators - 18 for the backbone and 4 for the wheels. This section briefly summarizes the key design features of the robot.

a) Robot backbone: The backbone is composed of a serially repeated line of 18 Dynamixel XM540 servo motors mounted on custom bent stainless steel brackets, alternating pitch and yaw joints as shown in Fig. 2-A and C. Each module includes one motor, one bracket, and a custom carrier board that has input and output ports for power and communication.

b) Head and tail: A RealSense camera is installed at the head. In addition, the head and tail modules are equipped with a pair of wheels, each of which is actuated

independently by a Dynamixel XM540. To accommodate these sensors and wheels, the head and tail modules are designed to be wider than the backbone modules. The wider wheel base helps the robot's stability during vertical motion.

c) Wheels: To increase traction on edges and corrugated pipes, the robot is equipped with a variant of a passively transformable single-part (PaTS) wheel [20] (Fig. 2-B). This wheel maintains its rolling efficiency while introducing the ability to morph into a long claw-like structure when exposed to point loads, which indicates a level of roughness that benefits from these protrusions. This has been found to help the robot grip the top edge of an obstacle such as a box, stair tread, or an edge like that of a pipe, and help it leverage itself onto the upper surface. We modified the original PaTS wheel design to be robust for the loads and added grousers to increase traction in the corrugated sections of the pipes.

d) Anisotropic bumper and skin: Each of the segments includes a sloped bumper that protects the carrier board and the servo during operations. The slope of the bumper faces towards the direction of travel to minimize catching of edges on terrain or obstacles. The backside of the bumpers has a lip that is designed to catch on ridges and points to arrest any slipping backward, aiding the robot in maintaining forward progression. Together, this design exhibits a directional anisotropic behavior. The bumpers also provide attachment points for an external rubber skin, which protects the system from debris. This skin allows quick access to the internal components by opening the seam or disconnecting the attachment points on the bumper.

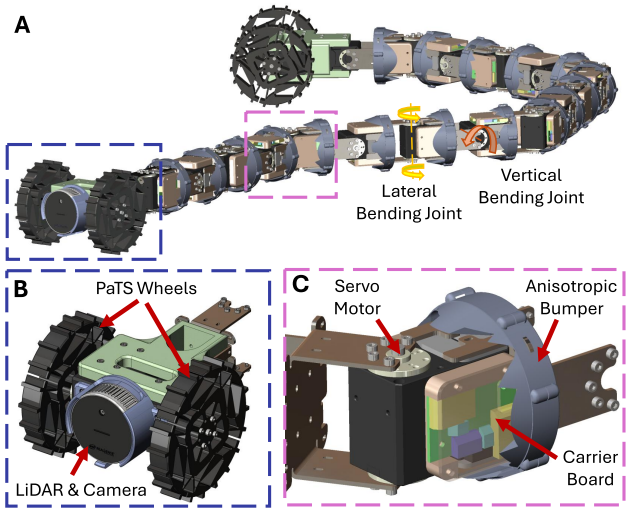


Fig. 2. The design of the Garden EELS robot. (A) Full robot assembly, including an 18-DOF spine with 4 driven wheels. (B) Head module featuring PaTS TPU wheels and Intel RealSense L515 LiDAR/Camera/IMU package. (C) Single body module including the Dynamixel XM540 servo motor, custom power and communication carrier board, TPU anisotropic circumferential bumper, and custom bent stainless steel brackets.

III. VERTICAL MOBILITY

This section presents our vertical mobility approach for clearing elevated access holes. We employ a *leader-follower locomotion*, in which each module of the snakebot roughly follows the path taken by the previous modules. The section concludes with a discussion on laboratory experiments and results.

A. Leader-Follower Locomotion

The methodology of the leader-follower controller can be broken into the following components: i) reference curve generation, ii) backbone control, iii) forward locomotion, iv) roll correction, and finally, v) vertical controller.

Reference Curve Generation: This step involves the generation of a trajectory for the robot to follow. Given a 2.5D elevation map of the environment and the robot's position within this map, a *reference curve* is generated by projecting a straight line from the head of the robot to a desired goal point. The height of this line is obtained by querying the elevation map, then smoothed to prevent any abrupt shift. To prevent rolling of the platform when clearing the step, a 90 degree turn to the left and then the right followed by a 2 m straight line is added. Finally, the backbone of the robot is appended to this curve. A visualization of this reference trajectory can be seen in Figure 3-A.

Backbone Control: In this step, the controller takes as input the reference curve and the relative position of the robot's tail with respect to the length of the curve, represented by $l \in [0, 1]$. The controller computes the desired $\mathbb{R}^3$ positions of the vertical and horizontal motors to place them on the curve, assuming the roll of the robot is zero and the path will be piecewise planar. Then, the algorithm starts from the tail and iterates through each joint to compute the angle required to place the next joint on the path. Given a joint $\mathbf{J_i}$ and the following joint, $\mathbf{J_{i+1}}$, the desired angle of joint $\mathbf{J_i}$, θ_i, is computed as follows:

$$\theta_i = \begin{cases} \arctan(y, x) & \text{if horizontal motor} \\ \arctan(z, x) & \text{if vertical motor} \end{cases} \quad (1)$$

where (x, y, z) is the position of $\mathbf{J_{i+1}}$ in the frame of $\mathbf{J_i}$.

Forward Locomotion: To achieve forward motion along the curve, the robot's wheels at the head and tail are rotated while the robot body undulates to produce forward force using friction against the contacting surface. The offset of the vertical motor positions from the desired positions is computed based on a sinusoidal curve:

$$\psi_i = v_a * \sin(f_s i + f_t t), \quad (2)$$

where ψ_i is the vertical offset of the ith motor's position, v_a is the desired vertical amplitude, f_s is the spatial frequency of the motor list along the curve, f_t is the temporal frequency at which the motors move along the curve, and t is time.

Roll Correction: A major issue in snake-robot step climbing is the robot's susceptibility to rolling while its body is lifted. Our controller corrects for roll by adding a curve to the horizontal motors. This is currently done through a simple

control command to set the first or last four horizontal motors to the input, ϕ, depending on whether the robot is ascending or descending the step.

Vertical controller: Finally, the vertical controller takes inputs from a human operator using a gamepad and orchestrates all of the controllers described above to clear large step heights. First, the human operator loads an a priori grid map of the environment and places the robot's position on the map. The user then defines a goal position for the robot to travel. The controller generates the reference curve and sends it to a control loop that computes both ϕ_i and ψ_i, and aggregates them to compute the joint positions for all motors given l and t, which are incremented or decremented based on a linear velocity command from the gamepad. The same command also moves the wheels forward and backward. Finally, the roll correction value is mapped to a linear axis on the gamepad control, allowing the user to lean the front or back of the robot during traversal.

B. Experimental Setups

We constructed a replica of the actual access hole in pit spaces, using authentic hardware, as shown in Fig 3-B. The key architectural features of interest include corrugated stainless steel access tunnels connecting adjacent pit spaces, and 40 cm-wide stainless steel U-handles anchored to rebar, which are installed for the ease of humans to traverse through the tunnel. Our replica is made of a four-section stackable 2x4 frame faced with HardiBacker cement boards and a concrete skim coat. This design allows for adjusting the tunnel height at two levels: 40 cm and 60 cm. By inserting concrete slabs between the sections, it can be raised to a greater height. The tunnel, measuring 60 cm in diameter and 75 cm in length, is secured with ratchet straps to maintain a screw-free interior.

C. Lab Experiments

The goal of the lab tests was to identify the robot's ability to overcome vertical steps. For each trial, the robot was initially placed directly in front of and centered on the vertical step. It then used the leader-follower locomotion to climb up, go through the elevated pipe, and climb down on the floor on the other side. Completing this sequence without external interference was classified as a successful trial, which was tested across variable height settings of 60, 76, and 92 centimeters for 5 trials each. Across these heights, we used two primary metrics. The first was the success rate, calculated as the number of successful trials divided by the number of trials. Secondly, the time taken from start to finish of only successful trials at that height was averaged.

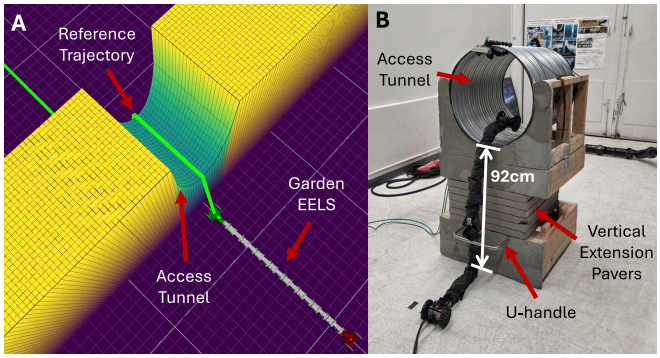


Fig. 3. Basic experimental setup and planning. (A) The generated reference trajectory based on input access tunnel height. (B) Obstacle setup including access tunnel, U-handle, and adjustable height concrete structure.

TABLE I
LAB RESULTS FOR VERTICAL LOCOMOTION AT VARIABLE HEIGHTS

Step Height	Success Rate	Avg. Time (s)
60 cm	100% (5/5)	367
76 cm	60% (3/5)	434
92 cm	20% (1/5)	454

The results are presented in Table I, which shows that Garden EELS is capable of navigating all of the height settings. As expected, the robot is more likely to succeed and takes less time on average for lower step heights. The recorded movie of the successful run with a 92 cm step height is included in the supplementary movie. The decreased success against taller obstacles is primarily due to the difficulty of maintaining stability in an upright pose when climbing up the step. The main balance of the robot came from the wheel base on the head and the tail, and instability increased as less of the robot was supported on the ground, yielding lower success rates as noted in Table I. When the robot did tip over, the trial was considered a failure. Manual control was added to yaw the head and tail live via remote control, which did increase the balance success rate, but human operation alone did not compensate fully. Feedback stability control would substantially increase the success rate, which is our future work.

While the leader-follower locomotion was executed fully autonomously, we added two manually controlled behaviors during the test to enhance the success rate. First, the rotation direction of the wheels was reversed when going through the U-shaped handles below the tunnel. We found that this behavior helped the robot's forward movement by pushing down the handle on the opposite side of the wall. Second, the roll-correction scheme described in Section III-A was deployed and was crucial for climbing the 76-cm and 92-cm steps. Automatically discovering and executing such maneuvers is a topic for future work, which would require algorithms that are aware of the physical interaction with the robot and the environment, such as model-based reinforcement learning.

IV. PLANAR LOCOMOTION WITH REINFORCEMENT LEARNING

A. RL Approach Overview

Next, we describe the RL-based controller used for horizontal planar locomotion. The policy receives commanded forward and angular body-frame velocities and generates joint motions to track these commands. Rather than producing arbitrary high-dimensional joint trajectories, the pol-

icy modulates a structured sinusoidal backbone curve that captures the dominant mode of undulatory locomotion. The controller is trained entirely in simulation and subsequently deployed on hardware without modification, demonstrating successful sim-to-real transfer under purely proprioceptive feedback.

We formulate horizontal locomotion for a biaxial, high-DOF snake robot as a reinforcement learning (RL) control problem. The objective is to learn a policy that maps commanded body-frame forward and angular velocities, illustrated in Fig. 4, together with proprioceptive observations, to parameters governing spatiotemporal joint trajectories. The policy does not observe global position or velocity in the world frame. Instead, it must infer appropriate gait modulation using only commanded velocities and joint feedback.

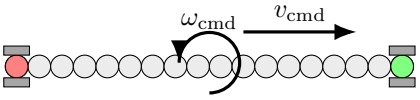


Fig. 4. Top-down schematic of a snake robot approximated by 18 spherical links, with commanded forward velocity v_{cmd} and angular velocity ω_{cmd}.

Unlike torque-level policies that directly output high-dimensional joint commands, we explicitly embed locomotion structure into the action space. As demonstrated in [22] and [23], many snake gaits are fundamentally sinusoidal; often, coherent body waves are necessary for generating propulsion and curvature. By constraining the policy to operate within a structured sinusoidal space, we reduce the dimensionality of the search space while preserving the essential mechanics of undulatory locomotion.

To reduce control dimensionality while preserving locomotion expressivity, the policy outputs parameters governing a single spatiotemporal sinusoidal backbone. The desired angle of the k-th joint is defined as

$$q(k, s) = a \sin\left(s + 2\pi\nu\frac{k}{N} + \phi\right), \tag{3}$$

where $k \in \{0, \ldots, N-1\}$ indexes joints, N is the number of joints, ν is spatial frequency, ϕ is a constant phase offset, and a is the amplitude produced by the policy network.

The phase variable evolves according to

$$s \leftarrow (s + 2\pi f dt) \bmod 2\pi, \tag{4}$$

following [24], where f is temporal frequency and dt is the control timestep. This representation captures the dominant mode of planar snake locomotion while eliminating redundant degrees of freedom associated with independent joint control. Rather than discovering arbitrary coordination patterns, the policy modulates physically meaningful gait parameters governing wave magnitude and propagation.

B. Unicycle-Integrated Pose Tracking Reward

A key challenge in learning planar velocity tracking for undulatory systems is reward formulation. A standard velocity-tracking reward penalizes instantaneous deviations,

$$r_{\mathrm{vel}} = -\|v - v_{\mathrm{cmd}}\|_2^2 - \|\omega - \omega_{\mathrm{cmd}}\|_2^2. \tag{5}$$

However, instantaneous body velocities oscillate due to internal wave propagation even when net displacement matches the commanded motion. Velocity-based rewards therefore introduce temporal noise. In practice, we found that a working policy could not be developed from this reward formulation.

To address this, we utilized a command-integrated pose tracking reward. Rather than comparing instantaneous velocities, we first integrate the commanded velocities over a short horizon using unicycle dynamics,

$$\dot{x} = v_{\mathrm{cmd}} \cos\theta, \quad \dot{y} = v_{\mathrm{cmd}} \sin\theta, \quad \dot{\theta} = \omega_{\mathrm{cmd}}, \tag{6}$$

producing a desired pose (x_d, y_d, θ_d). The reward then penalizes deviation between the actual pose and this integrated reference,

$$r_{\mathrm{pose}} = -\|(x, y, \theta) - (x_d, y_d, \theta_d)\|_2^2. \tag{7}$$

This formulation converts instantaneous velocity tracking into short-horizon trajectory tracking. By integrating commands before comparison, the reward suppresses high-frequency oscillations induced by body-wave propagation and collision while preserving accumulated displacement.

To our knowledge, prior RL formulations for snake locomotion rely primarily on constant velocities or forward-progress objectives [25]. The proposed command-integrated reward explicitly accounts for the oscillatory nature of undulatory locomotion and provides a principled mechanism for stabilizing velocity-tracking learning.

C. Ablation of Structured Parameterization

In addition to reward design, we introduce a sign-coupled structure in the action parameterization. For planar locomotion, the direction of wave propagation must be consistent with the sign of the commanded forward and angular velocities. Reversing the commanded velocity requires reversing the effective direction of the backbone wave to produce motion in the opposite direction.

When amplitude and spatial frequency signs are learned freely, the policy must simultaneously discover both magnitude and direction of wave propagation. This increases learning difficulty and can result in inconsistent turning behavior. We therefore explicitly couple the sign of the amplitude and spatial frequency to the sign of the commanded velocities, while allowing the policy to learn only their magnitudes.

We ablate three components of this structured parameterization. Each configuration is denoted by a 3-bit tag $b_1 b_2 b_3$, where:

- b_1: **Amplitude constraint.** If enabled, the policy predicts a nonnegative magnitude $\tilde{a}$, and its sign is determined by the commanded velocities. If disabled, the policy predicts a signed amplitude.
- b_2: **Temporal frequency constraint.** If enabled, the temporal frequency f is fixed. If disabled, f is learned within a bounded range.
- b_3: **Spatial frequency constraint.** If enabled, the spatial frequency ν is fixed and its direction is determined by the commanded velocities. If disabled, ν is learned within a bounded signed range.

Table II reports convergence steps and unicycle integrated position tracking errors for each configuration. Fully unconstrained models (000) require substantially more environment interaction to converge than structured variants. Incorporating sign coupling and frequency constraints reduces convergence time by up to 40% relative to the unconstrained baseline.

TABLE II

STRUCTURED PARAMETERIZATION ABLATION. VALUES ARE MEAN ± STANDARD DEVIATION OVER THREE SEEDS.

$b_1b_2b_3$	**Steps** [10^6]	**RMSE**$_{xy}$ [10^{-3} m]	**RMSE**$_\theta$ [10^{-3} rad]
111	$\mathbf{5.2}$	32.097 ± 2.592	$\mathbf{31.717 \pm 1.346}$
110	8.6	27.789 ± 0.680	41.133 ± 0.833
101	8.6	28.622 ± 2.643	33.454 ± 1.914
011	$\mathbf{5.2}$	26.017 ± 2.065	42.670 ± 1.793
100	10.3	25.693 ± 1.884	40.981 ± 1.619
010	6.9	23.228 ± 1.285	40.192 ± 0.913
001	5.7	24.203 ± 1.323	38.048 ± 1.379
000	8.6	$\mathbf{23.082 \pm 0.738}$	40.269 ± 0.768

Although some partially structured variants achieve comparable or slightly lower final RMSE values in simulation (by thousandths of a meter), removing directional coupling leads to less consistent turning behavior during hardware deployment. We therefore adopt the fully structured configuration (111) on hardware to ensure smooth motion and interpretable wave directionality.

D. Simulation and Hardware Deployment

Simulation is performed in MuJoCo with a physics timestep $dt_{\text{sim}} = 0.005$ s and control rate $dt_{\text{ctrl}} = 0.02$ s. During training, commanded velocities are sampled uniformly from $v_{\text{cmd}} \in [-0.035, 0.035]$ m/s and $\omega_{\text{cmd}} \in [-0.070, 0.070]$ rad/s, which were observed to be well within the hardware limits during deployment. Commands are resampled every 250 control steps within each 1000-step episode.

Following training, the learned policy is deployed on the physical robot without modification. A hierarchical control architecture is implemented in which a user provides body-frame unicycle velocity commands via joystick, while the learned policy generates sinusoidal joint trajectories using only proprioceptive feedback. The structured action representation confines motion to a physically meaningful gait manifold, contributing to robust sim-to-real transfer.

V. INSPECTION

The Garden EELS robot features two capabilities for inspection: 3D mapping of the confined space and object detection. The 3D mapping capability is provided by SERPENT, a multi-sensor SLAM framework that features explicit covariance modeling in the iterative closest point (ICP) registration, allowing the optimizer to weight measurements correctly and avoid overconfidence in unobservable directions [26]. This approach enables robust 3D mapping in underground pit spaces with noisy measurements due to dust, multipath interference, and broadband light sources that are often used in construction sites and interfere with LiDAR. As for object detection, we trained a YOLOv8n model [27] to detect flaws in concrete, leaking pipes, and objects left in rooms. In addition, we developed a user interface using PyQt5 and PyVista [28] to show the 3D map and object detection results in real time.

VI. FIELD DEMONSTRATION

The field demonstration took place at Kajima Training Center (KTC) in Kawasaki, Japan. We used two pit spaces in the foundation of the facility, which are connected by an elevated access, as shown in Figure 1. The floors are made of concrete coated with dust, with a 60-cm square gravel drainage sump at the center of the pit spaces. While only the single test site was used, its dimensions are representative, as shown in Table III.

TABLE III

THE MEASUREMENTS OF THE KEY FEATURES IN UNDERGROUND PIT SPACES.

Feature	**Portfolio Range**	**Test site (KTC)**
Wall Thickness	50–150 cm	100 cm
Ceiling Height	100–200 cm	177 cm
Tunnel Height	40–60 cm	57 cm

A. Vertical Locomotion

Figure 5 shows a successful vertical locomotion run at the test site. The Garden EELS robot was initially placed on the floor of one of the pit spaces, in front of the access tunnel. Then the robot followed a commanded path with the leader-follower gait to climb up into the tunnel, traverse it, and climb down onto the floor of the other pit space. We tested over 20 times with different initial poses, control parameter settings, and traverse directions, achieving an overall traverse success rate of $\sim$ 70%. The supplementary movie shows the complete traversal sequence. Among the two manual behaviors described in Section III-C, we only used the wheel direction control.

B. Horizontal Locomotion

The Garden EELS robot successfully executed sustained planar locomotion using the trained RL policy under joystick control on both the concrete floor and the gravel drainage sump, as shown in Figure 6 and the supplemental movie. The policy was solely trained in simulation with no on-site fine-tuning.

C. Inspection

We successfully demonstrated the 3D mapping capability using the SERPENT algorithm in the KTC pit spaces, as shown in Figure 7 and the supplementary movie. The object detection model was trained on a dataset that contains 29,599 images across five classes and is split 8:1:1 for training, validation, and testing. The model achieves strong mAP50 scores at the field site: 0.834 (crack), 0.881 (honeycomb), 0.956 (tools), 0.944 (helmet), and 0.892 (person), demonstrating reliable detection in real-world conditions.

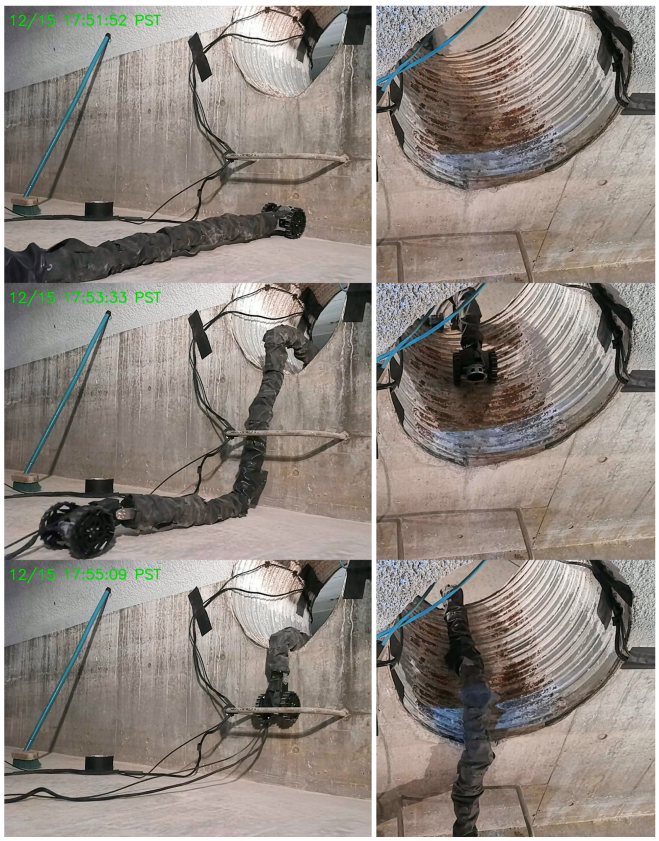


Fig. 5. A successful vertical traversal through the elevated access tunnel in the underground pit spaces at Kajima Technical Center.

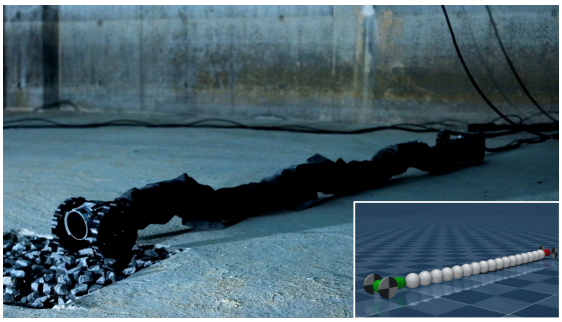


Fig. 6. Hardware deployment of the sinusoidal PPO policy for planar locomotion in the underground pit spaces at Kajima Technical Center and the simulated system in MuJoCo.

VII. CONCLUSIONS

We presented a new lightweight snake-like robot with a minimalist design for robotically inspecting the pit spaces in building foundations, which features three capabilities. First, using a leader-follower locomotion, it can go over large steps up to 92 cm in height. Notably, unlike existing robots that have comparable step-climbing abilities, our robot does not use proximity/contact sensing, nor is it equipped with wheels at every module. Second, we developed a reinforcement learning framework that is optimized for learning serpentine locomotion. A planar locomotion policy was trained in a MuJoCo simulator and successfully transferred to the hardware. Third, the robot builds upon existing SLAM and object detection algorithms for inspection. All three capabilities are successfully demonstrated in lab experiments, and field demonstrations took place in actual pit spaces at Kajima Technical Center.

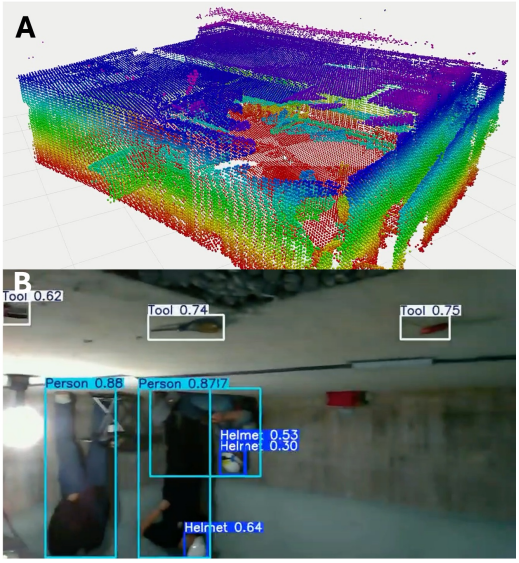


Fig. 7. Demonstration of the inspection capability in a pit space at Kajima Technical Center. (A) 3D voxel map created by the SERPENT algorithm. (B) Object detection by YOLOv8n.

ACKNOWLEDGMENT

The research was funded by Kajima Corporation and carried out at the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration (80NM0018D0004), as well as at the California Institute of Technology and Kyungpook National University.

We deeply thank Mr. Dai Ohama, Mr. Yasufumi Matsui, Mr. Eiji Hashimoto, Mr. Yuji Kawabata, Mr. Keisuke Hagiwara, Mr. Yuji Doi, and many other employees of Kajima Corporation who supported this work.

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